DSC5001: Statistical Machine Learning I

Tutorial 10: CART – Classification, Regression, and Pruning

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Task 1: Order Screening

A: account age (months). M: order amount (USD). $Y = 1$: chargeback.

ID	A	M	Y	ID	A	M	Y	ID	A	M	Y
1	1	40	1	5	6	110	0	9	12	180	0
2	2	100	1	6	7	70	0	10	14	130	0
3	3	80	1	7	8	220	1	11	16	90	1
4	4	150	0	8	9	60	0	12	20	240	1

Compare $A \leq 3.5$ and $M \leq 200$ using $G = 1 - \sum_k \widehat{p}_k^2$.

(a) Find each root split's weighted Gini reduction. Choose the better one.

(b) Within $A > 3.5$, find the *local* gain for $M \leq 200$.

(c) Using these two splits, give three leaf rules, $\widehat{P}(Y = 1)$, and predictions.

(d) For $A > 3.5,\ M \leq 200$, compare majority vote with the minimum-cost decision (FP cost 1, FN cost 8).

Solution 1(a): Account Age

The root has $(n_0, n_1) = (6, 6)$, so $G(P) = 1/2$.

$A \leq 3.5$	(n_0, n_1)	Size	Gini
Left	$(0, 3)$	3	0
Right	$(6, 3)$	9	$2(6/9)(3/9) = 4/9$

Weight the child impurities by their sizes:

$$G_A = \frac{3}{12}(0) + \frac{9}{12}\left(\frac{4}{9}\right) = \boxed{\frac{1}{3}}.$$

$$\boxed{\Delta G_A = \frac{1}{2} - \frac{1}{3} = \frac{1}{6} \approx 0.167.}$$

Solution 1(a): Order Amount

For the other proposed rule, $M \leq 200$:

	(n_0, n_1)	Size	Gini
Left	$(6, 4)$	10	$2(6/10)(4/10) = 0.48$
Right	$(0, 2)$	2	0

$$G_M = \frac{10}{12}(0.48) + \frac{2}{12}(0) = \boxed{0.40}.$$

$$\Delta G_M = 0.50 - 0.40 = \boxed{0.10}.$$

Choice: $\boxed{A \leq 3.5}$ has the larger gain of these two candidates.

Solution 1(b): Second Split

Within $A > 3.5$, the counts are $(6, 3)$:

$$G(P) = 2\left(\frac{6}{9}\right)\left(\frac{3}{9}\right) = \frac{4}{9}.$$

The proposed $M \leq 200$ split gives:

Child	(n_0, n_1)	Size	Gini
$M \leq 200$	$(6, 1)$	7	$12/49$
$M > 200$	$(0, 2)$	2	0

$$G_{\text{child}} = \frac{7}{9}\frac{12}{49} + \frac{2}{9}(0) = \frac{4}{21}.$$

$$\boxed{\Delta G_{\text{local}} = \frac{4}{9} - \frac{4}{21} = \frac{16}{63} \approx 0.254.}$$

Solution 1(c): Leaf Predictions

Each leaf's probability uses only its own observations.

Rule	$\widehat{P}(Y=1)$	$\widehat{Y}$
$A \leq 3.5$	$3/3 = 1$	1
$A > 3.5,\ M \leq 200$	$1/7$	0
$A > 3.5,\ M > 200$	$2/2 = 1$	1

At equal error costs, the predicted label is the majority class.

Key point: A leaf's probability is not calculated from all 12 orders.

Solution 1(d): Unequal Costs

In the middle leaf, $p = \widehat{P}(Y = 1) = 1/7$.

Decision	Expected cost per order
Predict 1	$C(1) = 1(1-p) = 6/7$
Predict 0	$C(0) = 8p = 8/7$

$$\boxed{C(1) < C(0) \quad \Longrightarrow \quad \widehat{Y} = 1.}$$

Equivalently, the cost-sensitive probability threshold is

$$\boxed{t = \frac{1}{1+8} = \frac{1}{9} < \frac{1}{7} = p.}$$

Majority voting predicts 0; unequal costs favor 1.

Task 2: Urgent Tickets

$Y = 1$: urgent ticket. Parent counts: $(n_0, n_1) = (14, 6)$.

Child	n_0	n_1	Size
Left L	9	1	10
Right R	5	5	10

Let $p_g = n_{1,g}/n_g$, $w_g = n_g/n_P$, and $G(g) = 2p_g(1 - p_g)$.

$$\Delta G = G(P) - w_L G(L) - w_R G(R).$$

(a) Compute ΔG. Does the number of majority-vote training mistakes decrease?

(b) Prove $\Delta G = 2w_L w_R (p_L - p_R)^2$.

(c) With FP cost 1 and FN cost 4, compare minimum total training cost before and after splitting.

(d) Minimum Gini gain is 0.10 and minimum leaf size is 6. Is the split allowed?

Solution 2(a): Gini Gain

The class-one proportions are 0.30, 0.10, and 0.50.

Node	p_1	$G = 2p_1(1 - p_1)$
Parent	0.30	0.42
Left	0.10	0.18
Right	0.50	0.50

The two children each contain 10 tickets:

$$G_{\text{child}} = \frac{1}{2}(0.18) + \frac{1}{2}(0.50) = \boxed{0.34}.$$

$$\boxed{\Delta G = 0.42 - 0.34 = 0.08.}$$

Solution 2(a): Training Errors

A leaf's fewest classification mistakes equal its minority-class count.

Tree	Error calculation	Total
Unsplit	$\min(14, 6)$	6
Split	$\min(9, 1) + \min(5, 5)$	$1 + 5 = 6$

$$\boxed{\Delta G = 0.08 > 0, \qquad \Delta\text{Errors} = 0.}$$

Insight: A split can improve Gini without reducing training mistakes. The right child is tied: either label makes 5 errors.

Solution 2(b): Gini Identity

The parent proportion is $p_P = w_L p_L + w_R p_R$, with $w_L + w_R = 1$.
Using $G(p) = 2p(1-p)$, first cancel the linear terms:

$$\begin{aligned}\Delta G &= 2p_P(1-p_P) - 2w_L p_L(1-p_L) - 2w_R p_R(1-p_R) \\ &= 2\big(w_L p_L^2 + w_R p_R^2 - p_P^2\big).\end{aligned}$$

Now substitute $p_P = w_L p_L + w_R p_R$:

$$\boxed{\Delta G = 2w_L w_R (p_L - p_R)^2 \geq 0.}$$

Here $w_L = w_R = 1/2$ and $(p_L, p_R) = (0.1, 0.5)$:

$$\boxed{\Delta G = 2(1/2)^2 (0.4)^2 = 0.08.}$$

Solution 2(b): The Key Algebra

Expanding the square in the preceding step gives

$$\begin{aligned} & w_L p_L^2 + w_R p_R^2 - (w_L p_L + w_R p_R)^2 \\ &= w_L(1 - w_L) p_L^2 + w_R(1 - w_R) p_R^2 - 2 w_L w_R p_L p_R \\ &= w_L w_R (p_L^2 - 2 p_L p_R + p_R^2) \\ &= \boxed{w_L w_R (p_L - p_R)^2}. \end{aligned}$$

For nonempty children, $w_L, w_R > 0$.

$$\boxed{\Delta G > 0 \quad \text{if and only if} \quad p_L \neq p_R.}$$

Insight: Purity improves when the two children have different class proportions.

Solution 2(c): Decision Costs

FP cost 1, FN cost 4; predict class 1 when $p_1 > 1/5$.

Node	p_1	Decision	Training cost
Parent $(14, 6)$	.30	1	$14(1) = 14$
Left $(9, 1)$	.10	0	$1(4) = 4$
Right $(5, 5)$	.50	1	$5(1) = 5$

$$\boxed{C_{\text{before}} = 14, \qquad C_{\text{after}} = 4 + 5 = 9.}$$

Insight: Cost-sensitive training loss falls even though majority-vote mistakes are unchanged.

Solution 2(d): Stopping Rules

Both conditions must hold:

$$\underbrace{n_L = n_R = 10 \geq 6}_{\text{minimum leaf size satisfied}}$$

$$\underbrace{\Delta G = 0.08 < 0.10}_{\text{minimum Gini gain not met}}.$$

Therefore

$$\boxed{\text{The split is not allowed.}}$$

Stopping is based on the prescribed Gini criterion, not the change in cost-sensitive training loss.

Task 3: Bike-Share Travel Time

Predict trip time y (minutes) from distance x (km):

x	1	2	3	4	5	6	7	8
y	6	7	8	9	17	18	19	22

A regression-tree leaf predicts its mean. Splits minimize within-leaf RSS.

(a) Find the root-node mean and RSS.

(b) Compare total child RSS for $s = 2.5, 4.5, 6.5$. Give the best two-leaf predictions.

(c) Prove the identity below and verify it at the selected cut:

$$RSS_P - RSS_L - RSS_R = \frac{n_L n_R}{n_P}(\bar{y}_L - \bar{y}_R)^2.$$

(d) For validation trips $(3.5, 10)$ and $(6.5, 20)$, compare the unsplit and two-leaf MSEs without refitting.

Solution 3(a): Root RSS

The eight responses give

$$\sum_i y_i = 106, \qquad \sum_i y_i^2 = 1688.$$

The unsplit tree predicts

$$\boxed{\bar{y} = 106/8 = 13.25.}$$

Using $RSS = \sum_i y_i^2 - (\sum_i y_i)^2/n$,

$$\begin{aligned} RSS_P &= 1688 - \frac{106^2}{8} \\ &= \boxed{283.5}. \end{aligned}$$

Solution 3(b): Candidate Cuts

Each candidate predicts its child sample means.

s	$\bar{y}_L$	$\bar{y}_R$	RSS_L	RSS_R	Total RSS
2.5	6.5	15.5	0.5	161.5	162
4.5	7.5	19	5	14	**19**
6.5	65/6	20.5	833/6	4.5	430/3

The best of these three thresholds is $\boxed{s = 4.5}$:

$$\boxed{\widehat{f}(x) = \begin{cases} 7.5, & x \le 4.5, \\ 19, & x > 4.5. \end{cases}}$$

Solution 3(c): Centering RSS

For either child $g \in \{L, R\}$, decompose a residual as

$$y_i - \bar{y}_P = (y_i - \bar{y}_g) + (\bar{y}_g - \bar{y}_P).$$

Squaring and summing within child g gives

$$\sum_{i \in g} (y_i - \bar{y}_P)^2 = RSS_g + n_g(\bar{y}_g - \bar{y}_P)^2,$$

because the cross term contains

$$\sum_{i \in g} (y_i - \bar{y}_g) = 0.$$

Summing over both children,

$$\boxed{RSS_P - RSS_L - RSS_R = n_L(\bar{y}_L - \bar{y}_P)^2 + n_R(\bar{y}_R - \bar{y}_P)^2.}$$

Solution 3(c): Between-Group Gain

Since

$$\bar{y}_P = \frac{n_L \bar{y}_L + n_R \bar{y}_R}{n_P},$$

we have

$$\bar{y}_L - \bar{y}_P = \frac{n_R}{n_P}(\bar{y}_L - \bar{y}_R),$$

$$\bar{y}_R - \bar{y}_P = -\frac{n_L}{n_P}(\bar{y}_L - \bar{y}_R).$$

Substitution into the previous slide yields

$$\boxed{RSS_P - RSS_L - RSS_R = \frac{n_L n_R}{n_P}(\bar{y}_L - \bar{y}_R)^2.}$$

Solution 3(c): Numerical Check

At $s = 4.5$:

$$n_L = n_R = 4, \qquad \bar{y}_L = 7.5, \qquad \bar{y}_R = 19.$$

The formula gives

$$\begin{aligned} RSS_P - RSS_L - RSS_R &= \frac{4 \cdot 4}{8}(7.5 - 19)^2 \\ &= 2(11.5)^2 \\ &= \boxed{264.5}. \end{aligned}$$

Direct comparison confirms

$$\boxed{283.5 - 19 = 264.5.}$$

Insight: Splitting reduces RSS when the two child means differ.

Solution 3(d): Validation MSE

For validation trips $(3.5, 10)$ and $(6.5, 20)$:

Model	$\widehat{y}(3.5)$	$\widehat{y}(6.5)$
Root only	13.25	13.25
Two leaves	7.5	19

$$MSE_{\text{root}} = \frac{(10 - 13.25)^2 + (20 - 13.25)^2}{2} = \boxed{28.0625}.$$

$$MSE_{\text{tree}} = \frac{(10 - 7.5)^2 + (20 - 19)^2}{2} = \boxed{3.625}.$$

The two-leaf tree wins on these two validation observations.

Task 4: Crop Monitoring

$Y = 1$: infested, $Y = 0$: healthy. Root branches A and B each have two leaves.

Leaf	A_1	A_2	B_1	B_2
(n_0, n_1)	$(10, 0)$	$(2, 8)$	$(7, 3)$	$(4, 6)$

Each leaf predicts its majority class. Let $R(T)$ be the number of training mistakes and $|T|$ the number of leaves. Use $J_\alpha(T) = R(T) + \alpha|T|$.

(a) Find $(R(T), |T|)$ for the full tree, pruning only A, pruning only B, and collapsing the root.

(b) For A, B, and the root, calculate $\alpha_t = [R(t) - R(T_t)]/(|T_t| - 1)$. Here $R(t)$ is the loss after collapsing node t. Which prunes first?

(c) Find the next root-pruning breakpoint. Select the best tree at $\alpha = 1, 2.5, 4$.

(d) CV errors for these α values are $0.18, 0.12, 0.21$. Select α and explain the role of the test set.

Solution 4(a): Tree Losses

A leaf's minimum training mistakes equal its minority-class count.

Tree	Collapsed counts	$R(T)$	$\lvert T\rvert$
Full	–	$0+2+3+4=9$	4
Prune A	$(12,8)$	$8+3+4=15$	3
Prune B	$(11,9)$	$0+2+9=11$	3
Root only	$(23,17)$	17	1

$$\boxed{(R,|T|)=(9,4),\ (15,3),\ (11,3),\ (17,1).}$$

Solution 4(b): Branch Penalties

The weakest-link penalty is

$$\boxed{\alpha_t = \frac{R(t) - R(T_t)}{|T_t| - 1}.}$$

Branch A: collapse (A_1, A_2) into one leaf:

$$\alpha_A = \frac{8 - (0 + 2)}{2 - 1} = \boxed{6}.$$

Branch B: collapse (B_1, B_2) into one leaf:

$$\alpha_B = \frac{9 - (3 + 4)}{2 - 1} = \boxed{2}.$$

The first candidate is $\alpha_B = 2$, but the root must also be checked.

Solution 4(b): Root Penalty

The full tree has $R(T) = 9$ and 4 leaves.
Collapsing the root creates one leaf with 17 mistakes:

$$\alpha_{\text{root}} = \frac{17 - 9}{4 - 1} = \boxed{\frac{8}{3} \approx 2.667}.$$

Compare all internal nodes:

$$\boxed{\alpha_B = 2 < \alpha_{\text{root}} = 8/3 < \alpha_A = 6.}$$

First prune: branch $\boxed{B}$.

Solution 4(c): Next Breakpoint

After pruning B, the remaining tree has 3 leaves and 11 errors.
Replacing it with the root-only leaf would give 17 errors and 1 leaf:

$$\boxed{\alpha_{\text{next}} = \frac{17 - 11}{3 - 1} = 3.}$$

The relevant objectives are

$$J_\alpha(T_4) = 9 + 4\alpha,$$
$$J_\alpha(T_3) = 11 + 3\alpha,$$
$$J_\alpha(T_1) = 17 + \alpha.$$

Pruning only A gives $15 + 3\alpha$, always worse than pruning B with the same number of leaves.

Solution 4(c): Tree Comparison

Evaluate the three candidate trees from the pruning sequence:

α	Full T_4	Prune B: T_3	Root T_1	Minimum
1	13	14	18	Full
2.5	19	18.5	19.5	Prune B
4	25	23	21	Root only

α	1	2.5	4
leaves selected	4	3	1

Solution 4(d): CV Selection

The provided CV errors are

α	1	2.5	4
$\mathrm{CV}(\alpha)$	0.18	**0.12**	0.21

The minimum occurs at

$$\boxed{\widehat{\alpha} = 2.5 \quad \Longrightarrow \quad \text{prune branch } B.}$$

Choose α using cross-validation on the training data. Use the untouched final test set only to evaluate the completed model.

Tutorial 10: CART splitting, prediction, and pruning

Questions and discussion